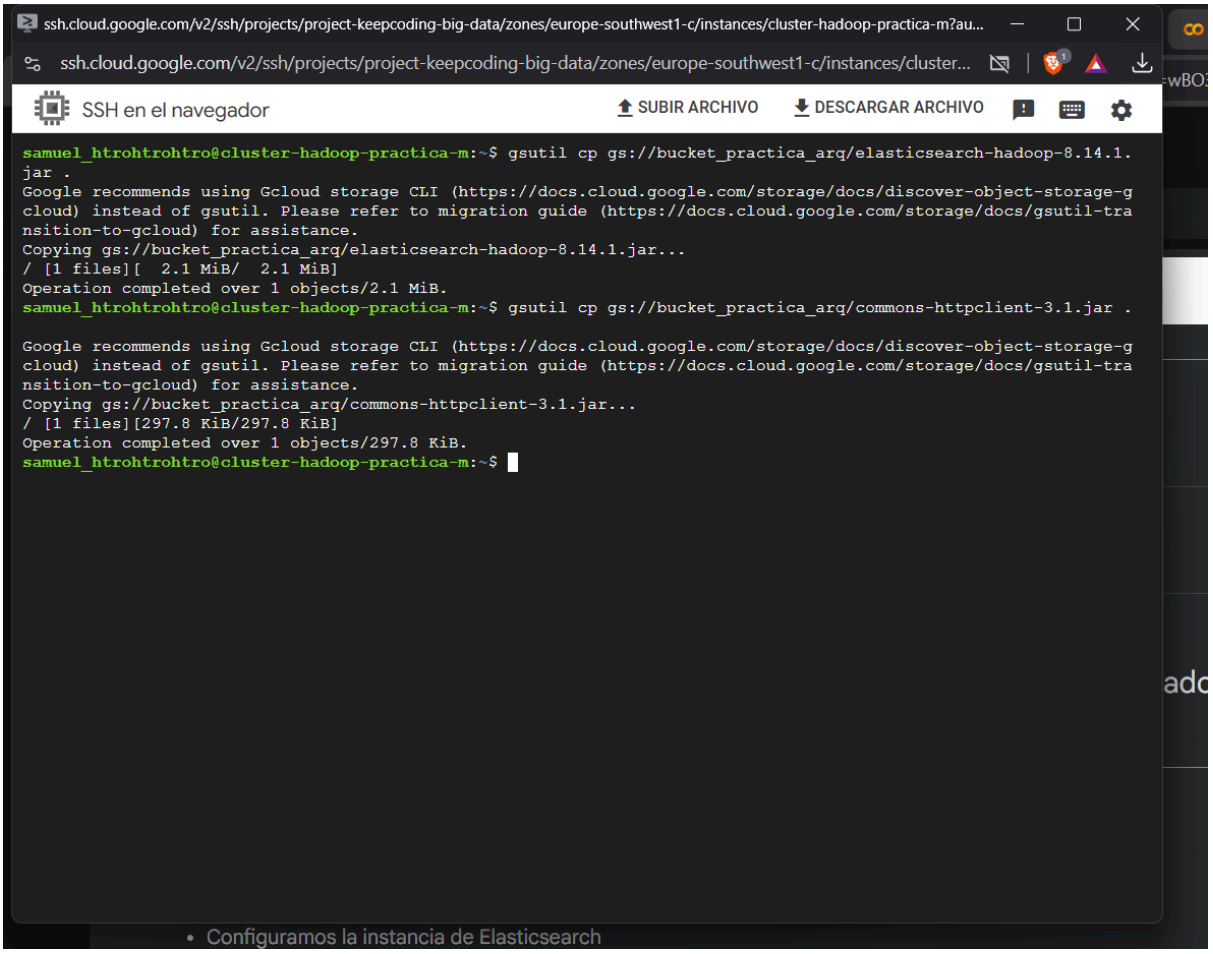
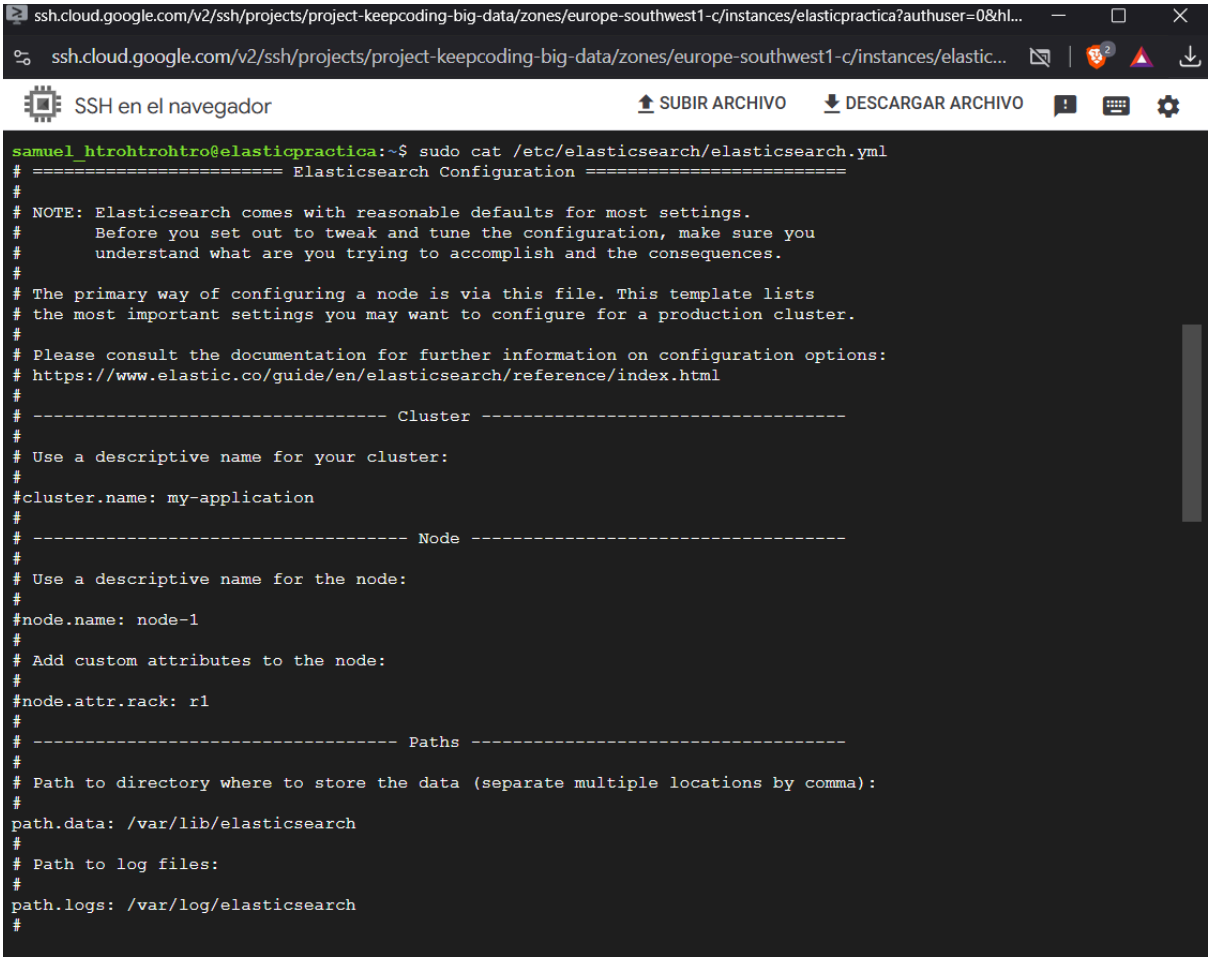
Parte 1: Captura de pantalla de la consola SSH del cluster Hadoop una vez finalizada la configuración y carga.



Parte 2: Captura de pantalla de la consola del server Elastic donde se vea la configuración de elastic, desde 'Enable security features' hasta el final (el fichero elasticsearch.yml) abierta.



ssh.cloud.google.com/v2/ssh/projects/project-keepcoding-big-data/zones/europe-southwest1-c/instances/elasticpractica?authuser=0&hl...

ssh.cloud.google.com/v2/ssh/projects/project-keepcoding-big-data/zones/europe-southwest1-c/instances/elastic...

SSH en el navegador SUBIR ARCHIVO DESCARGAR ARCHIVO

```
path.logs: /var/log/elasticsearch
#
# ---------------------------------- Memory -----------------------------------
#
# Lock the memory on startup:
#
#bootstrap.memory_lock: true
#
# Make sure that the heap size is set to about half the memory available
# on the system and that the owner of the process is allowed to use this
# limit.
#
# Elasticsearch performs poorly when the system is swapping the memory.
#
# ---------------------------------- Network ----------------------------------
#
# By default Elasticsearch is only accessible on localhost. Set a different
# address here to expose this node on the network:
#
#network.host: 192.168.0.1
#
# By default Elasticsearch listens for HTTP traffic on the first free port it
# finds starting at 9200. Set a specific HTTP port here:
#
#http.port: 9200
#
# For more information, consult the network module documentation.
#
# --------------------------------- Discovery ----------------------------------
#
# Pass an initial list of hosts to perform discovery when this node is started:
# The default list of hosts is ["127.0.0.1", "[::1]"]
#
#discovery.seed_hosts: ["host1", "host2"]
#
# Bootstrap the cluster using an initial set of master-eligible nodes:
#
#cluster.initial_master_nodes: ["node-1", "node-2"]
#
```

ssh.cloud.google.com/v2/ssh/projects/project-keepcoding-big-data/zones/europe-southwest1-c/instances/elasticpractica?authuser=0&hl...

ssh.cloud.google.com/v2/ssh/projects/project-keepcoding-big-data/zones/europe-southwest1-c/instances/elastic...

SSH en el navegador SUBIR ARCHIVO DESCARGAR ARCHIVO

```
#cluster.initial_master_nodes: ["node-1", "node-2"]
#
# For more information, consult the discovery and cluster formation module documentation.
#
# ---------------------------------- Various -----------------------------------
#
# Allow wildcard deletion of indices:
#
#action.destructive_requires_name: false

#----------------------- BEGIN SECURITY AUTO CONFIGURATION -----------------------
#
# The following settings, TLS certificates, and keys have been automatically
# generated to configure Elasticsearch security features on 22-03-2026 15:55:12
#
# ----------------------------------------------------------------------------

# Enable security features
xpack.security.enabled: false

xpack.security.enrollment.enabled: true

# Enable encryption for HTTP API client connections, such as Kibana, Logstash, and Agents
xpack.security.http.ssl:
  enabled: false
  keystore.path: certs/http.p12

# Enable encryption and mutual authentication between cluster nodes
xpack.security.transport.ssl:
  enabled: true
  verification_mode: certificate
  keystore.path: certs/transport.p12
  truststore.path: certs/transport.p12
# Create a new cluster with the current node only
# Additional nodes can still join the cluster later
cluster.initial_master_nodes: ["elasticpractica"]

# Allow HTTP API connections from anywhere
# Connections are encrypted and require user authentication
```

```
# Additional nodes can still join the cluster later
cluster.initial_master_nodes: ["elasticpractica"]

# Allow HTTP API connections from anywhere
# Connections are encrypted and require user authentication
http.host: 0.0.0.0

# Allow other nodes to join the cluster from anywhere
# Connections are encrypted and mutually authenticated
#transport.host: 0.0.0.0

#----------------------- END SECURITY AUTO CONFIGURATION -------------------------
samuel_htrohtrohtro@elasticpractica:~$
```

Parte 3: Captura de pantalla del proceso de configuración en Cluster Hadoop de Conexión con ES completo.

ssh.cloud.google.com/v2/ssh/projects/project-keepcoding-big-data/zones/europe-southwest1-c/instances/cluster-hadoop-practica-m?au...

ssh.cloud.google.com/v2/ssh/projects/project-keepcoding-big-data/zones/europe-southwest1-c/instances/cluster...

SSH en el navegador SUBIR ARCHIVO DESCARGAR ARCHIVO

```
  Memory usage: 40%               IPv4 address for ens4: 10.204.0.8
  Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

9 updates can be applied immediately.
9 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

7 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.


Last login: Sun Mar 22 15:25:02 2026 from 35.235.246.32
samuel_htrohtrohtro@cluster-hadoop-practica-m:~$ sudo sed -i '$d' /etc/hive/conf.dist/hive-site.xml
samuel_htrohtrohtro@cluster-hadoop-practica-m:~$ sudo sed -i '$a \  <property>\n    <name>es.nodes</name>\n    <
value>34.175.132.179</value>\n  </property>\n' /etc/hive/conf.dist/hive-site.xml
samuel_htrohtrohtro@cluster-hadoop-practica-m:~$ sudo sed -i '$a \  <property>\n    <name>es.port</name>\n    <v
alue>9200</value>\n  </property>\n' /etc/hive/conf.dist/hive-site.xml
samuel_htrohtrohtro@cluster-hadoop-practica-m:~$ sudo sed -i '$a \  <property>\n    <name>es.nodes.wan.only</nam
e>\n    <value>true</value>\n  </property>\n' /etc/hive/conf.dist/hive-site.xml
samuel_htrohtrohtro@cluster-hadoop-practica-m:~$ sudo sed -i '$a \  <property>\n    <name>hive.aux.jars.path</na
me>\n    <value>/usr/lib/hive/lib/elasticsearch-hadoop-8.14.1.jar,/usr/lib/hive/lib/commons-httpclient-3.1.jar</v
alue>\n  </property>\n</configuration>' /etc/hive/conf.dist/hive-site.xml
samuel_htrohtrohtro@cluster-hadoop-practica-m:~$ sudo cp elasticsearch-hadoop-8.14.1.jar /usr/lib/hive/lib/
samuel_htrohtrohtro@cluster-hadoop-practica-m:~$ sudo cp commons-httpclient-3.1.jar /usr/lib/hive/lib/
samuel_htrohtrohtro@cluster-hadoop-practica-m:~$ sudo systemctl restart hive
Failed to restart hive.service: Unit hive.service not found.
samuel_htrohtrohtro@cluster-hadoop-practica-m:~$ sudo systemctl restart hive-metastore
samuel_htrohtrohtro@cluster-hadoop-practica-m:~$ sudo systemctl restart hive-server2
samuel_htrohtrohtro@cluster-hadoop-practica-m:~$
```

Parte 4: Captura de pantalla de la consola del cluster Hadoop con el resultado de la consulta.

ssh.cloud.google.com/v2/ssh/projects/project-keepcoding-big-data/zones/europe-southwest1-c/instances/cluster-hadoop-practica-m?au...

ssh.cloud.google.com/v2/ssh/projects/project-keepcoding-big-data/zones/europe-southwest1-c/instances/cluster...

SSH en el navegador SUBIR ARCHIVO DESCARGAR ARCHIVO

```
curl: (6) Could not resolve host: IP-SERVER-ELASTIC
samuel_htrohtrohtro@cluster-hadoop-practica-m:~$ curl -X GET "http://34.175.132.179:9200/alumnos/_search?pretty"
{
  "took" : 30,
  "timed_out" : false,
  "_shards" : {
    "total" : 1,
    "successful" : 1,
    "skipped" : 0,
    "failed" : 0
  },
  "hits" : {
    "total" : {
      "value" : 6,
      "relation" : "eq"
    },
    "max_score" : 1.0,
    "hits" : [
      {
        "_index" : "alumnos",
        "_id" : "6",
        "_score" : 1.0,
        "_source" : {
          "title" : "New Document",
          "content" : "This is a new document for the master class",
          "tag" : [
            "general",
            "testing"
          ]
        }
      },
      {
        "_index" : "alumnos",
        "_id" : "3",
        "_score" : 1.0,
        "_source" : {
          "id" : 3,
          "name" : "Carlos",
          "last_name" : "González"
```

ssh.cloud.google.com/v2/ssh/projects/project-keepcoding-big-data/zones/europe-southwest1-c/instances/cluster-hadoop-practica-m?au...

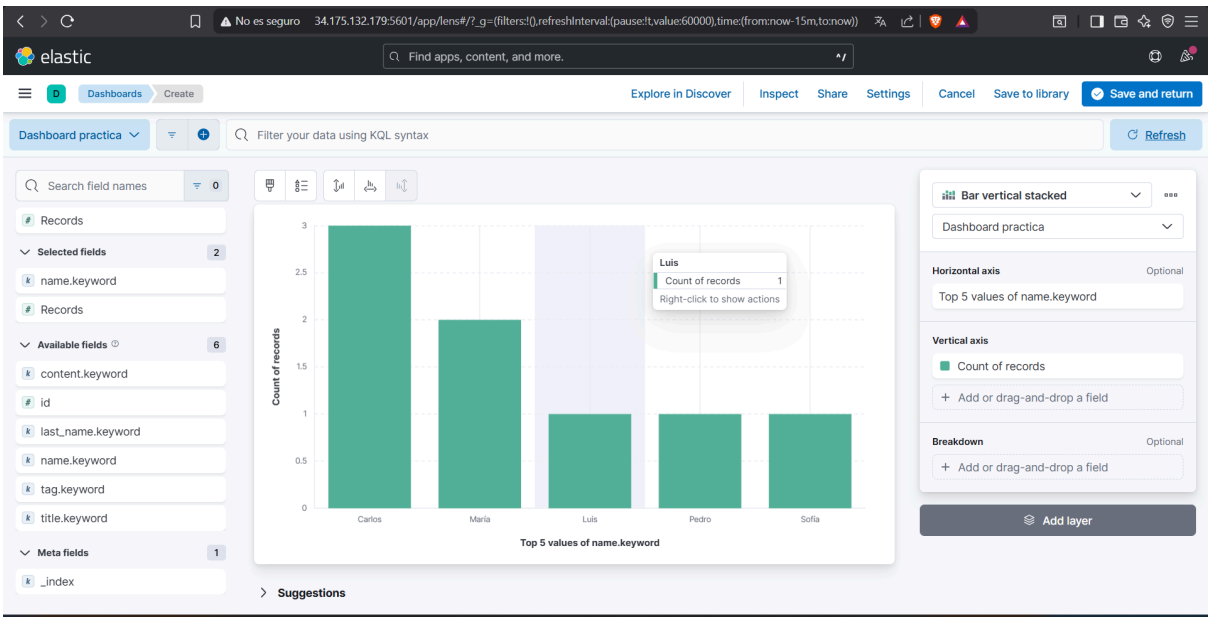
ssh.cloud.google.com/v2/ssh/projects/project-keepcoding-big-data/zones/europe-southwest1-c/instances/cluster...

SSH en el navegador SUBIR ARCHIVO DESCARGAR ARCHIVO


```
          "name" : "Carlos",
          "last_name" : "González"
        }
      },
      {
        "_index" : "alumnos",
        "_id" : "4",
        "_score" : 1.0,
        "_source" : {
          "id" : 4,
          "name" : "María",
          "last_name" : "López"
        }
      },
      {
        "_index" : "alumnos",
        "_id" : "5",
        "_score" : 1.0,
        "_source" : {
          "id" : 5,
          "name" : "Luis",
          "last_name" : "Martínez"
        }
      },
      {
        "_index" : "alumnos",
        "_id" : "7",
        "_score" : 1.0,
        "_source" : {
          "id" : 7,
          "name" : "Sofía",
          "last_name" : "Ramírez"
        }
      },
      {
        "_index" : "alumnos",
        "_id" : "8",
        "_score" : 1.0,
        "_source" : {
```



```
        }
      },
      {
        "_index" : "alumnos",
        "_id" : "8",
        "_score" : 1.0,
        "_source" : {
          "id" : 8,
          "name" : "Pedro",
          "last_name" : "Hernández"
        }
      }
    ]
  }
}
samuel_htrohtrohtro@cluster-hadoop-practica-m:~$
```

Parte 5: Captura de pantalla de la consola de Kibana con alguna visualización sencilla.

Para que el dashboard tuviese más sentido añadí más alumnos con el mismo nombre, e hice un dashboard que contase los alumnos con el mismo nombre